Lab_Syn_2.0

You have to synthesize Up_Dn_Counter.v file using Design Compiler shell and generate Technology Dependent gate level netlists in Verilog format using standard cell libraries (typical, worst, best) inside std_cells folder:-

- scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db

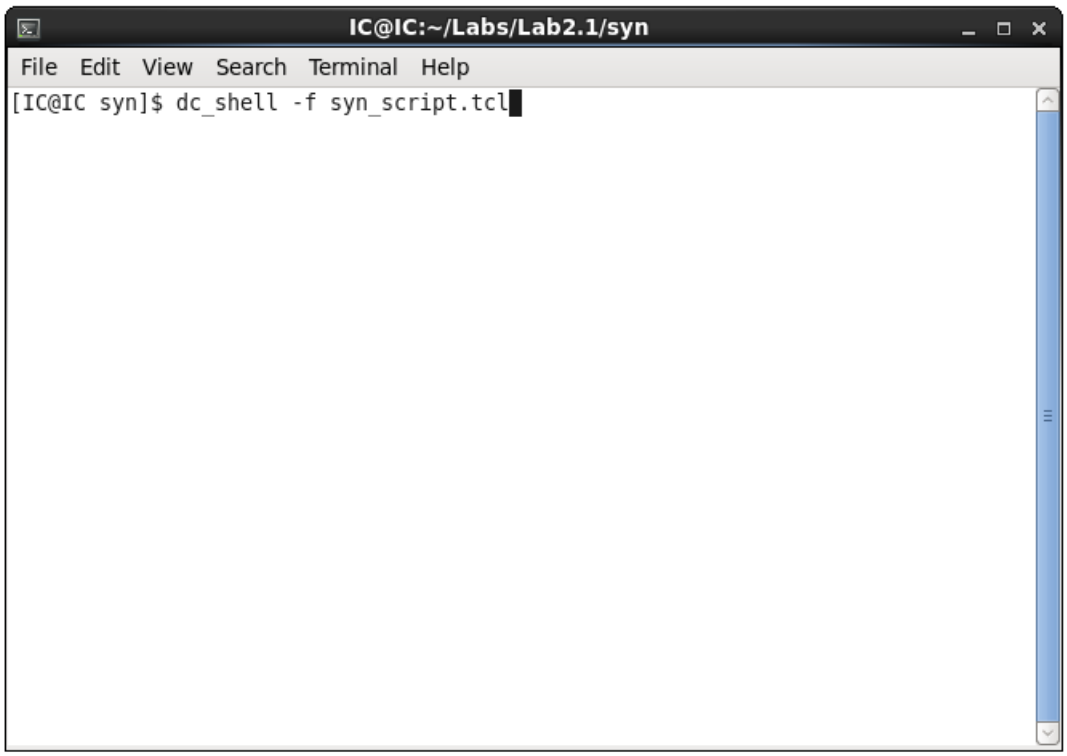
Steps: -

1. Copy Lab_Syn_2.0 folder from your computer into the virtual machine inside Labs directory
2. Create a script with name syn_script.tcl and write the following commands
   1) Add the std cell libraries path to the search path
   2) Add the rtl files path to the search path
   3) Define the values of target_library and link_library variables to load standard cell libraries (typical)
   4) Read the Up_Dn_Counter.v file using read_file command
   5) Link the design & tech files using link command
   6) Map & Optimize the design using compile command
   7) Generate the mapped gate level netlist using write_file command

Open Save Undo

syn.log | syn_script.tcl | syn_script.tcl

```
1
2 ################## Design Compiler Library Files #################
3
4 lappend search_path /home/IC/Labs/Lab_Syn_2.0/std_cells
5 lappend search_path /home/IC/Labs/Lab_Syn_2.0/rtl
6
7 set TTLIB "scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db"
8
9 ## Standard Cell libraries
10 set target_library [list $TTLIB]
11
12 ## Standard Cell & Hard Macros libraries
13 set link_library [list * $TTLIB]
14
15 ######################## Reading RTL Files ######################
16
17 read_file -format verilog Up_Dn_Counter.v
18
19 link
20
21 ###################### Mapping and optimization ################
22
23 compile
24
25 ##################################################################
26 # Write out Design after initial compile
27 ##################################################################
28
29 write_file -format verilog -output Up_Dn_Counter_mapped.v
30
```

3. Open the terminal inside path /home/IC/Labs/Lab_Syn_2.0/syn

4. Run the script through design compiler shell using the following command in the terminal and press Enter: -
   - dc_shell -f syn_script.tcl